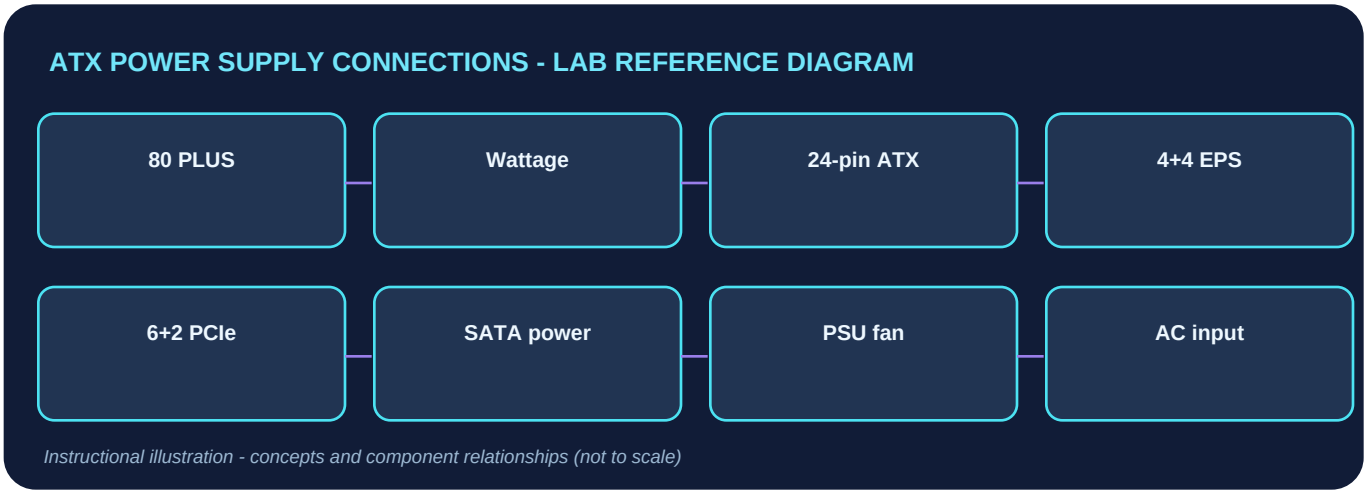
3.1.2 Install a Power Supply

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Identify PSU wiring and power requirements.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Compare calculated load to PSU rated output.
2. Disconnect AC and observe electrostatic precautions.
3. Install PSU and connect mainboard, CPU and drive power.
4. Verify startup and stable voltages using safe methods.

Troubleshooting and safe practice

• No POST: verify 24-pin and CPU EPS connectors.

• Avoid opening PSU enclosure; stored energy can be hazardous.

3.1.2 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
80 PLUS	A key term in atx power supply connections; identify its purpose, where it is used, and how to verify it.
Wattage	A key term in atx power supply connections; identify its purpose, where it is used, and how to verify it.
24-pin ATX	Primary motherboard power connector.
4+4 EPS	CPU power connector commonly grouped as 8 pins.
6+2 PCIe	A key term in atx power supply connections; identify its purpose, where it is used, and how to verify it.
SATA power	Electrical power characteristic or component; assess safely with rated tools and correct connectors.
PSU fan	A key term in atx power supply connections; identify its purpose, where it is used, and how to verify it.
AC input	A key term in atx power supply connections; identify its purpose, where it is used, and how to verify it.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____